



Rhinoplus Arge Danışmanlık İmalat San.Tic. Ltd. Şti.

Bahçelievler Mah. 319 Cad. E Blok (Ankara Üni. Teknokent) 35 E-B13

Gölbaşı/ Ankara

[Http://Rhinoplus.com.tr](http://Rhinoplus.com.tr)

info@Rhinoplus.com.tr

(90) 312 585 02 70

Geomembrane Welding Robot

PLAWEL S7 Operating Instructions Manual (En)



PLAWEL is a geomembrane welding robot and mainly used for welding of geomembrane sheets in the application areas like waste storage places, ponds (artificial lakes), medical waste collection areas, treatment and irrigation pools, canals, dams etc. PLAWEL is the one of the best products in the market considering the weight and the price/quality ratio together.

General Properties:

Voltage	V	220
Power Consumption	W	2500
Temperature	°C	0 - 450
Travel speed	m/min	1.2-7
Material thickness	mm	0.5-3
Weight	Kg	9.5
Dimension	mm	330x240x250
Shipping Weight (With Metallic Case)	Kg	16

Temperature adjustment:

In the figure below the temperature unit (element + wedge) has been shown. The temperature can be assigned between 0-450 °C according to the membrane material and thickness. Also, the wedge the type of material is defining the type of wedge where for PVC membrane stainless steel wedge and for others copper wedge is used. If the same copper wedge is used for PVC, the wedge part will wear out and the cost of the problems arising from this issue is entirely at the customer's expense.

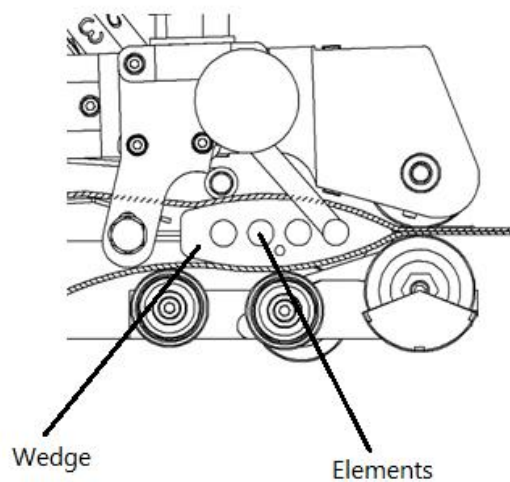


Fig1. Temperature unit on wedge

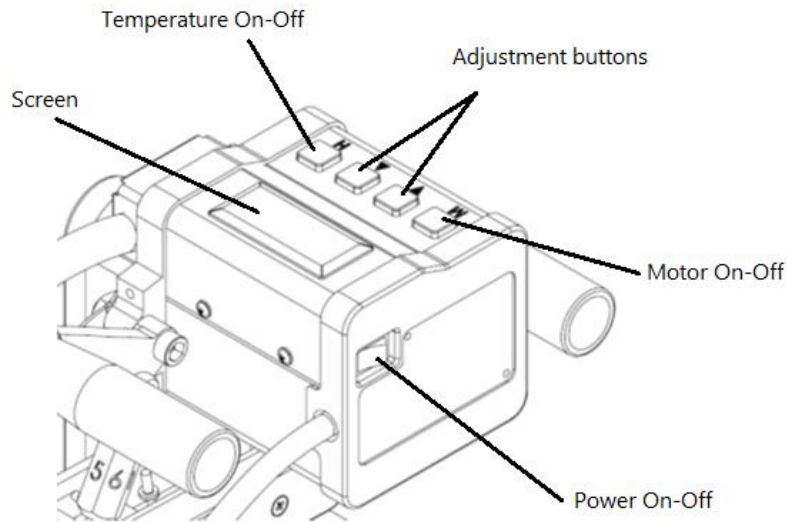


Fig2. Adjustment and On-Off buttons

To adjust the temperature, push the “H” button (Temperature On-Off) and one of the adjustment buttons (Δ or ∇) at the same time to enter to temperature adjustment menu. when the screen starts to blink and shows the temperature, you are in the adjustment menu, and you can leave your hand. While screen blinks you can hold Δ or ∇ buttons to increase or decrease the temperature. As soon as the desired temperature is reached just leave the buttons and push the “H” button again to save.

Warning: The wedge part is the hottest part of the machine, so do not touch the wedge with your bare hand and keep the wedge away from the materials that can burn quickly.

Pressure adjustment:

To adjust the welding pressure, there is an adjustment bolt which is designed to apply the forces 100 to 1000 N to the membranes. The bolt is designed to apply 100 N per number. Therefore, before starting to turn the bolt just hold down the pressure lever (Fig.3) and put two overlapped membrane between the rollers. Then start to turn the bolt until both upper and lower rollers get in contact with the membranes without applying any pressure. Write down the number you can read (Example, in the figure you can read number 4) and accept it as zero force. After this, any number in the bolt will apply excessive 100 N to the membranes.

Please be sure NOT to pass 10 turns which is equal to 1000 N. After few tryouts you will understand what the best pressure is considering the welding quality.

Note: It is advised to use 500 N for a membrane with 1.5 mm thickness as manufacturer suggestion.

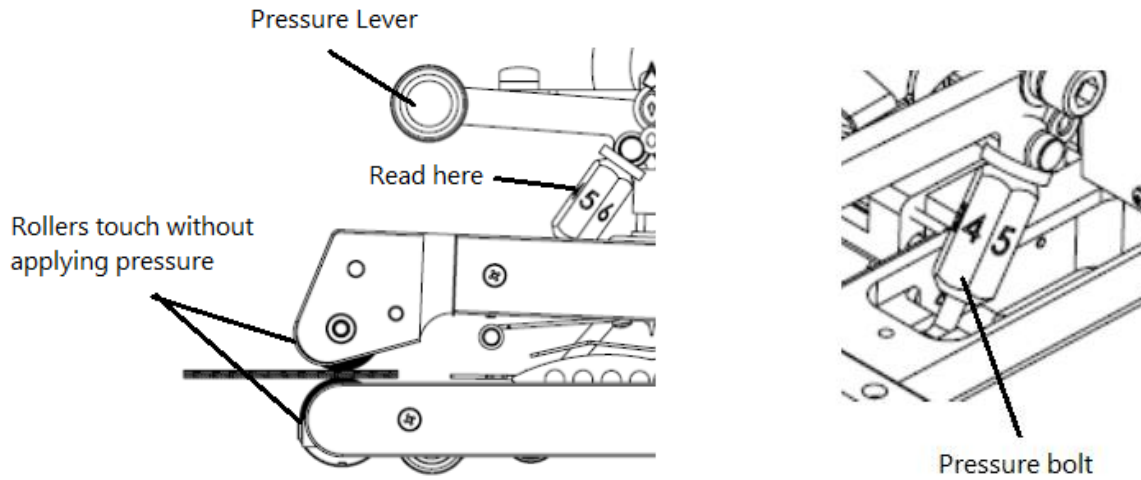


Fig 3. Pressure unit at position 4 showing 400 N

Warning: be careful not to get your fingers caught in the wedge area when you lower the pressure lever. Also, note that nothing except the geomembranes is trapped between the rolls (Figure 4) or the wedge.

Guiding roller adjustment:

The guiding rollers are used to make the membrane have maximum contact with the hot wedge so the welding quality increases.

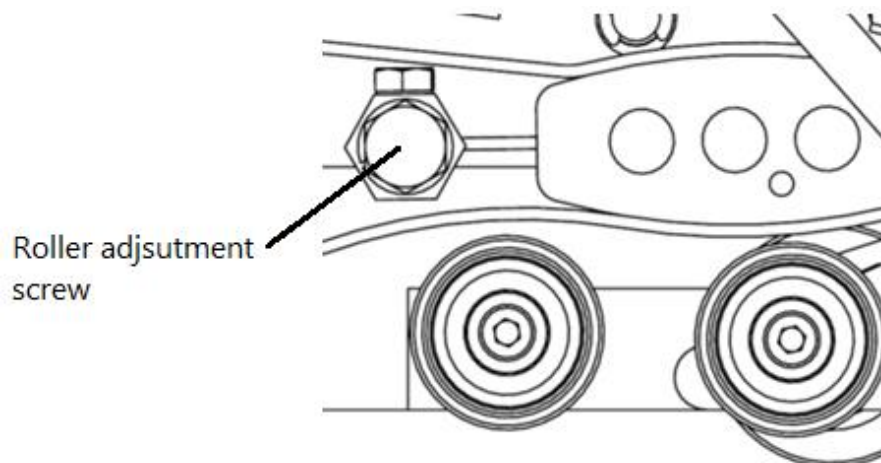


Fig.4. Roller adjusting with two different screws

If you have already adjusted the temperature, pressure and rollers, you need to do some arrangements before starting to weld.

- Clean the membranes and overlap them between 80 to 120 mm.
- The power supply should be at least 4 KW with a voltage stabilizer.
- Extension cord should be chosen as below,

230 V~	to 50m	3x1,5mm ² / 3x14 AWG
	to 100m	3x2,5mm ² / 3x12 AWG

- Be sure that the pressure Lever is NOT clamped.
- Adjust the welding speed using “M” and “Δ” or “▽” buttons.
- Be sure to clean the wedge and rollers before welding.
- Be sure to lubricate the rollers and check if they can turn freely.

How to weld:

1. Plug the cable into a 220 V electrical outlet. Make sure that the power supply has a ground connection. The responsibility for electric shock or any other problem due to lack of ground connection is at the customer's responsibility.
2. Press the "H" button so that the resistances start to heat up and reach the predefined temperature. You can see on the screen that the heating has started. Please do not touch the wedge and the metal parts around it with your bare hands.
3. Try to put the overlapped membranes as below into the space between wedge and rollers. One should go under the wedge and the other one over it. while placing the membranes, make sure that your hand is not caught or burned by touching the hot wedge.
4. The wedge position can be adjusted horizontally. Be sure that it is in the right end of the wedge slider considering Figure 5.
5. Now the wedge temperature and motor speed are both set, membranes are both in the place. You can now push the pressure lever down to clamp and start the welding.
6. The robot will automatically pull the membranes inside the pressure rollers and the melted surface of the membranes will get welded. At this stage, speed and temperature will be monitored online from the screen. Do not allow fingers or other foreign objects to be pulled between the rolls.

7. As soon as the robot reaches to the end of the welding route, open the pressure lever up to avoid the pressure rollers (Fig.5) sliding on each other.
8. Push back the wedge on slider and turn off the elements using “H” button and the motor with “M” button.
9. Do the vision test on the welding and if the quality is not satisfactory, decrease the speed using “M” and “V” buttons.

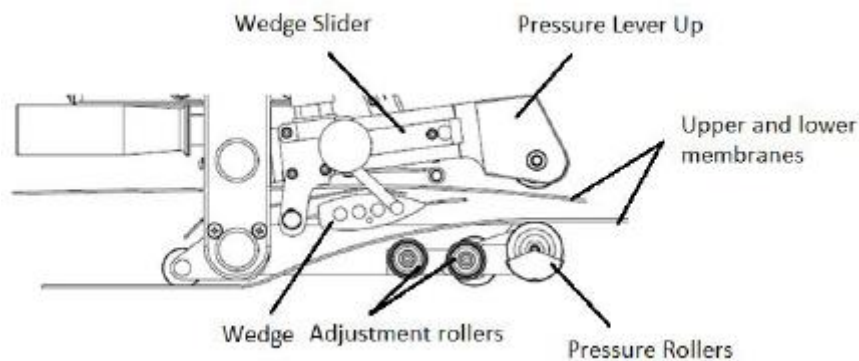


Fig.5. The way of putting membranes into the robot

Maintenance period:

- Check the geomembrane welding robot every six months after it is started to weld.
- Before any maintenance, pull out the plug from the socket.
- You can check the wedge is not worn out, the chain is not loosened, the teeth of the pressure rollers are not worn- out, the cable is not damaged, the rollers rotate normally, and the temperature is not dropped more than 70 degrees from the set point during the welding operation.
- When the cables are damaged, contact our technical service.
- Clean the wedge after the usage. Soft metal brushes can be used for this job. Do not use plastic cloth or brushes. Do not use any chemical liquid for the cleaning.
- Check the rollers and make sure that stones, mud, or any foreign objects are not stuck.
- If any part replacement is required, please contact our technical service, and do not try to replace it by yourself.
- Check the above-mentioned things every 100,000 m of welding, even if the robot is not reached to its six months operational age.
- Make sure that the wedge is not hot during the maintenance.

- Make sure that children do not touch or play with the robot during the maintenance.
- The serial number of the robot is written at the bottom of the device as a marking. The first four digits of this number indicate the year it was produced. The warranty of the device starts from the invoice date and has nothing to do with this date.

Warnings:



The hot wedge is the hottest part of the machine and can cause a fire. Ensure that the wedge does not come into contact with materials that can burn quickly and avoid placing near such materials.



Do NOT touch the hot wedge with bare hands. Be careful not to touch the hot wedge during the welding, after welding finished, during the placing of the membranes, or during the cleaning.



This device works with high voltage. To avoid electrical shock, be sure to check the robot before any use. This problem usually occurs in a wet environment. Your power supply must be properly grounded, otherwise the customer is responsible for the possible electric shocks.



Be careful not to get your fingers caught in the wedge area when you lower the pressure lever. Also, make sure that nothing foreign apart from the geomembranes is trapped between the rolls or the wedge. While the robot is running, do not put your fingers between the pressure rollers or the wedge and be careful the rollers do not pull any foreign objects into the area.



Be sure to use a voltage stabilizer to avoid electrical damage to the controller. If you are using a generator, pay attention to have the grounding connection and voltage stabilizer.



Do not touch or try to change the fuse of the robot. Also, do not attempt to change any bolts or parts of the robot. Contact our technical service in any case.



As the robot is not waterproof, it must be protected against the water or moisture. The responsibility of the robot not working due to these reasons or any electric shock during the operation is on the customer.

- Keep an eye on children during the use, the maintenance or the cleaning and make sure that they do not touch this robot.

- After your work is done, place the robot on a safe place or surface that will not be damaged by the hot wedge, and wait for the robot wedge to cool down to room temperature. After the device is cooled down, it can be placed in its box.
- Do not leave the device in an operation and have your eyes on it.
- Unplug the robot from the socket during the maintenance, cleaning or changing accessories.
- Use the original spare parts when any fails.
- When the cable of the robot is damaged, pull the plug from the socket and send the robot to the service.
- Do not try to repair the robot by yourself.



Fig.6. The warning for hot wedge and high voltage on the Plawel robot

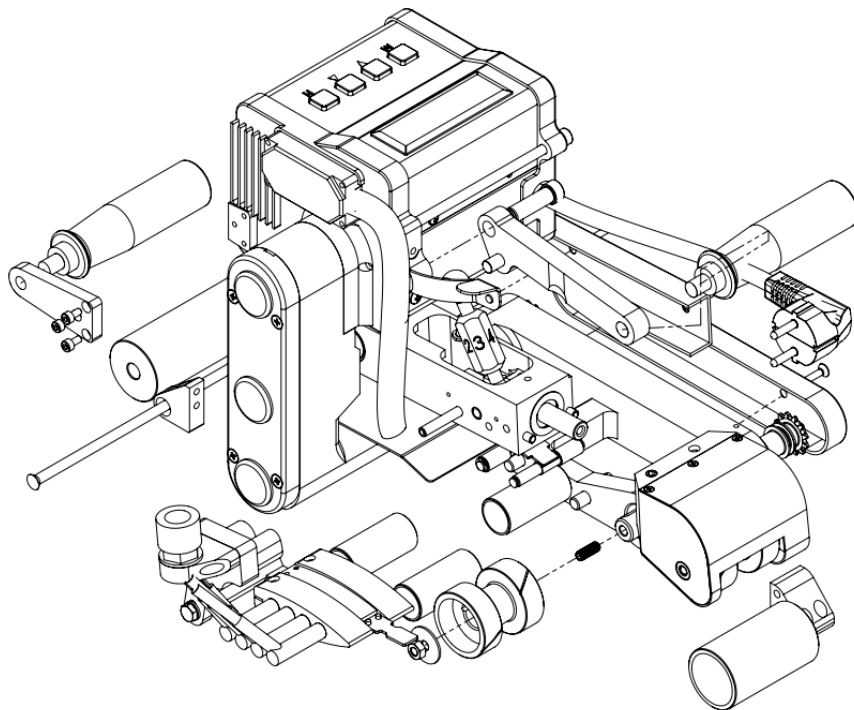


Fig.7. exploded view of Plawel S7 robot

Warranty:

Warranty period is one year from the invoice date.

Warranty start date:	Company stamp/signature:
Robot Serial No, <i>Plawel – S7:</i>	

A document without a stamp cannot be considered as a warranty document.

Out of guaranteed situations:

- Damage due to voltage fluctuation
- Damage due to wet environment
- Damage due to mechanical impact
- The parts which have finished their lifetime (like elements) are out of guarantee
- Any attempt to open the machine parts by the customer will end the guarantee
- Any damage due to misuse